SVN

Lecture: 01 date: 10/08/2025

Machine Learning and Regression Analysis

slide

Machine learning Applications & computation

Jahangir Alom

Book

Machine learning

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ML-Real-World Applications

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Machine learning

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CSE 4139

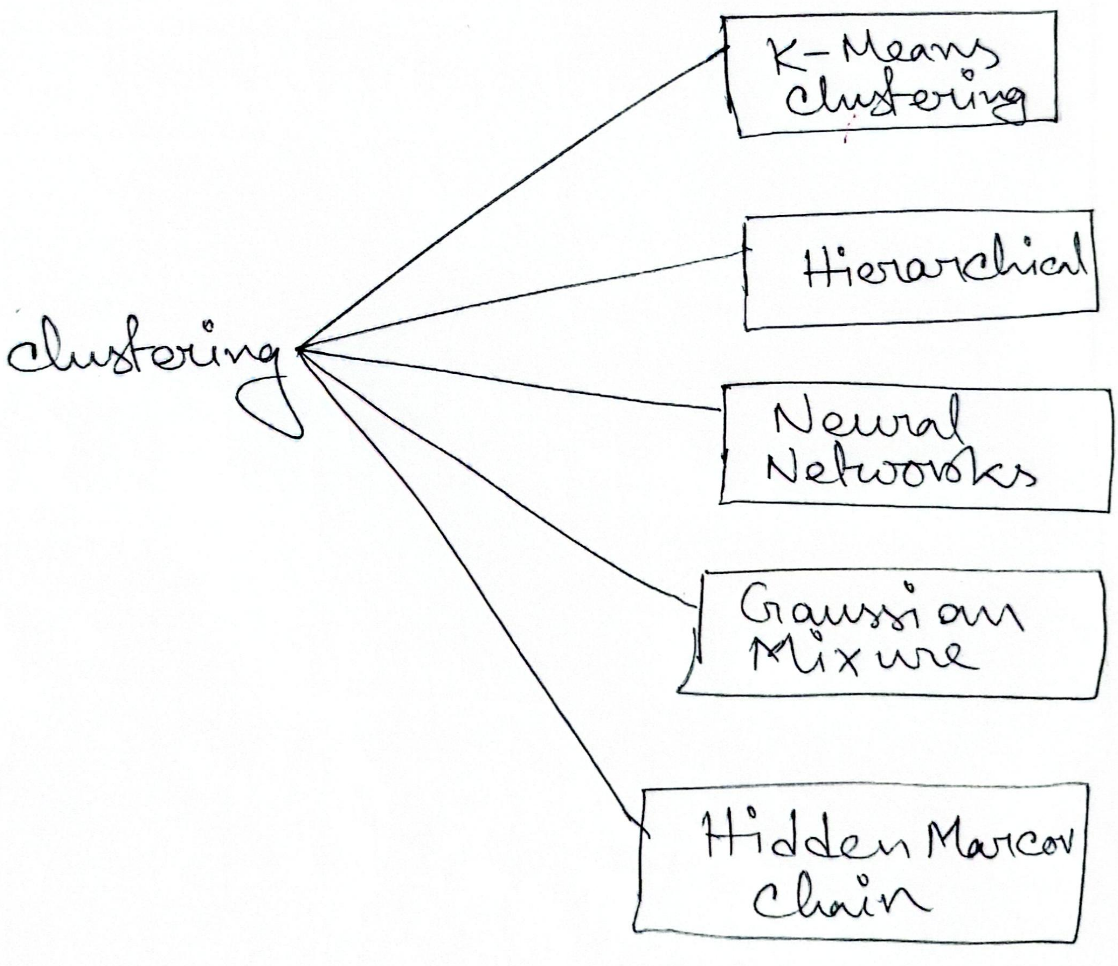
Machine Learning

Lecture-3 13.08.2025

⊞ Supervised Machine Learning:

⊞ Unsupervised ML :

Goal: Discover patterns or structure from unlabeled data.



কতগুলো cluster আছে সেটা বের করার জন্য total কতগুলো Data আছে তার Square Root বের করতে হবে। Ex: 15 data. ∴ cluster = $\sqrt{15}$
= 3/4

(*) LDA

Machine Learning & Deep Learning:

Semi-Supervised ML:

Goal: Use a mix of labeled and unlabeled data.

Key Idea: Labeled data is scarce, so unlabeled data helps boost performance.

ANN & Deep Learning Models:

Reinforcement Learning:

Learning through trial and error with rewards and penalties.

Goal: learn actions that maximize rewards through trial and errors.

⊞ SL, USL, RL :

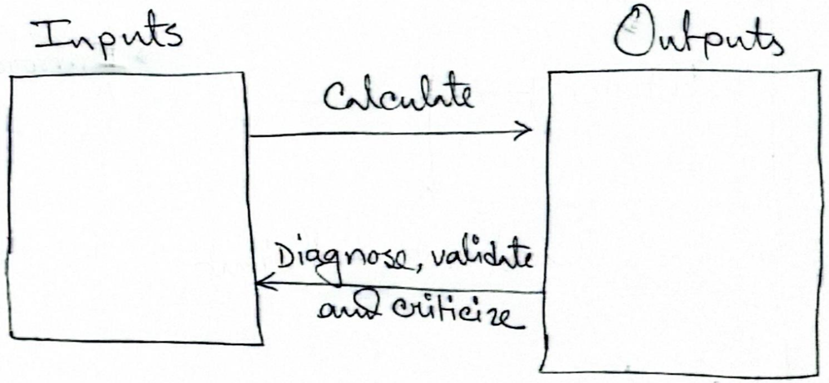
⊞ Ensemble Methods:

Combine multiple models

⊞ Time Complexity of Most Popular ML Algorithms:

** Regression Analysis **

⊞ Steps in Regression Analysis :



(**) Flowchart →

⊞ Various Classification of Regression Analysis:

⊞ Degree of Freedom (df):

Model specification:

Linear:

$$Y = \beta_0 + \beta_1 X_1 + \varepsilon$$

β_0 → Intercept

β_1 → Constant Coefficient

ε → Errors

Non-linear:

$$Y = \beta_0 + e^{\beta_1 X_1} + \varepsilon$$

If X is nonlinear:

$$Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \varepsilon$$

$$\Rightarrow Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \varepsilon$$

$$Y = \beta_0 + \beta_1 \ln X + \varepsilon$$

$$\Rightarrow Y = \beta_0 + \beta_1 X + \varepsilon$$

Model Fitting: Linear Regression:

Observations	X (Hours of Studied)	Y (Exam Score)
1	1	2
2	2	3
3	3	5
4	4	4
5	5	6

Fit a model:

$$Y = \beta_0 + \beta_1 X + \varepsilon$$

β_0 : intercept

β_1 : Slope

$$\bar{X} = (1+2+3+4+5)/5 = 3$$

$$\bar{Y} = (2+3+5+4+6)/5 = 4$$

$$\beta_1 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2} = \frac{9}{10} = 0.9$$

$$\beta_0 = \bar{Y} - \beta_1 \bar{X}$$

$$= 4 - 0.9 \times 3 = 4 - 2.7 = 1.3$$

$$Y = \beta_0 + \beta_1 X + \varepsilon$$

$$\boxed{Y = 1.3 + 0.9X + \varepsilon}$$

Calculate residuals for each point (optional): $e_i = Y_i - \hat{Y}$

Question: Criticize the Score of the regression Model

H.W: Rainfall Prediction Model using ML